Tahmid Hasan Sakib

Cookeville, Tennessee | tsakib42@tntech.edu | LinkedIn | GitHub | Google Scholar

PROFESSIONAL SUMMARY

Graduate researcher in autonomous-vehicle perception, secure ROS 2, and controlled cyber-physical validation. Combine scaled physical testbeds, CARLA simulation, experiment documentation, and research communication.

RESEARCH METHODS AND TOOLS

Methods: clean-baseline calibration, physical comparison groups, recorded-log replay, live fallback testing, detector and range evidence review, failure analysis.

Platforms and software: Python, C/C++, Linux, Git, Jetson TX2/AGX Orin, ROS 2/SROS2 on QCar2, CSI/RealSense cameras, 2D LiDAR, CARLA, YOLOv5n/YOLOv8.

RESEARCH EXPERIENCE

Graduate Research Assistant | Tennessee Technological University, Cookeville, TN | May 2024-Present

- Analyzed QCar1 clean and printed-sign physical runs using detector, LiDAR-range, and vehicle-stop evidence; documented configuration differences across groups.
- Traced QCar2 SROS2 provisioning and middleware trust boundaries in a controlled qualitative proof of concept.
- Compared PULSE monitor flags with recorded-log and live QCar2 outcomes; reviewed REDAP frame-level confidence and later E3/E4 control traces with collaborators.
- Mentored NSF REU undergraduates in summers 2025 and 2026 under faculty supervision; personally presented a poster at MECC 2025.

Hardware Engineer Intern | Robi Axiata Limited, Dhaka, Bangladesh | December 2019-January 2020

- Prototyped AVR/Arduino systems and simulated hardware and communication behavior with MATLAB, LTspice, and Cisco Packet Tracer.

SELECTED STUDIES

QCar1 physical adversarial perception

Across 30 clean runs and separate 18-run pre/post printed-sign groups, runs with a missed stop changed from 8/18 to 0/18; detector and runtime settings both changed.

PULSE physical monitoring and fallback

In 10 live black/white QCar2 runs, 18/20 patched encounters were flagged and all 20 stop sequences completed; stop-line position was not independently measured.

REDAP restoration-based defense

Offline analysis of 151 stored QCar2 frames found 57/151 attacked versus 110/151 restored frames at the 0.60 cutoff; later E3/E4 closed-loop outcomes were mixed.

EDUCATION

Tennessee Technological University | M.S., Electrical and Computer Engineering, expected December 2026 | GPA 4.00/4.00

Master's thesis defended; official degree conferral pending.

Military Institute of Science and Technology, Dhaka, Bangladesh | B.Sc., Computer Science and Engineering, August 2021

PUBLICATIONS AND PRESENTATIONS

Accepted: First-author IEEE Access QCar1 adversarial-perception article, 2026. Published: First-author IEEE MILCOM 2025 workshop and ACMI 2021 papers; coauthor IEEE SATC 2026 paper.

Presentation: Personally presented MECC 2025 extended-abstract poster, Pittsburgh, October 5-8, 2025. PULSE: submitted and under review, VTC 2027-Spring. REDAP: CVPR 2027 registered; full submission pending.